Considering more substantial changes to the paradigm

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Alternative Approach
backgroud

- We have been exploring under what conditions perceptions about the future could be a driver of business cycles.
- We have focused on Walrasian settings. Findings?
  1. Difficult: two challenges.
  2. Most promising: Departing for representation agent framework. Helpful in getting perceptions to increase desire to trade in both investment and consumption goods.
- What may we be missing by maintaining Walrasian framework
  1. Does not capture possibility of inefficiently large recessions (not taking a vacation).
  2. No real coordination problem (can be low demand, but not deficient demand ?)
  3. outcomes ex-post optimal

Alternative Approach

- We now examine departures from Walrasian framework which may involve coordination problems
- Steps
  1. Static Coordination games (Cooper-John).
     - Multipliers and Multiple Equilibrium - Contrast with Walrasian setting - Link with Global Games
  2. First look at dynamics (only expectations of current variables)
     - Adding capital (external effect due to incomplete markets or IRS) - Illustrate emergence of hysterisis versus limit cycle
  3. Second look at dynamics (with forward component)
     - IRS, local indeterminacy (Benhabib-Farmer) - Complementarities, limit cycles and multiple equilibrium (Diamond-Fudenberg)
  4. Extended example: Hayek-Keynes, limit cycles and local stability

- I players with payoffs function $V(e_j, \bar{e})$, where $\bar{e} = \sum_i \frac{e_i}{I}$
- social optimum FOC (objective: $\max \sum V(e_j, \bar{e})$) are given by

$$V_1(e_j, \bar{e}) + \sum_i V_2(e_j, \bar{e}) \frac{\partial \bar{e}}{\partial e_j} = V_1(e, e) + V_2(e, e) = 0$$

- SOC

$$V_{11} + 2V_{21} + V_{22} < 0$$

- we will assume that this SOC is satisfied and that $V_{11} < 0$

Individual decisions

- Individual level Problem: $Max_{e_j} V(e_j, \bar{e})$
- Two possibilities. Agents takes into account effect on aggregate

$$V_1(e_j, \bar{e}) + V_2(e_j, \bar{e}) \frac{\partial \bar{e}}{\partial e_j} = V_1(e_j, \bar{e}) + \frac{V_2(e_j, \bar{e})}{I} = 0$$

- Or, if player disregards his effect on aggregate (common assumption for macro-setups)

$$V_1(e_j, \bar{e}) = 0$$

- We will focus on second, but this has minor importance. When I is very large, the two become similar. Note, as I get larger, departure for social optimal conditions gets larger.

Best responses

- WE will want to allow $V(\cdot)$ to be affected by exogenous forces. This can be captured by including a parameter θ, ie, payoffs are of form $V(e_j, \bar{e}; \theta)$. This will allow for comparative static exercises
- Let $e_j(\bar{e}; \theta)$ represent the agents best response function, that is, $e_j(\bar{e}; \theta)$ is defined as the solution to

  $$V_1(e_j, \bar{e}) = 0$$

  for a given value of $\bar{e}$
- Equilibrium is easiest to describe in e-$\bar{e}$ space. Relationship can be positive, negative, non-linear...

SNE

- Symmetric Nash Equilibrium
- Assuming player disregards his effect on aggregate (common assumption for macro-setups), SNE is set of e such that

$$V_1(e, e; \theta) = 0$$

- or, set of e such that $e_j(\bar{e}; \theta) = \bar{e}$
- We will also assume that $V(\cdot)$ is such that an interior solution exists

Definitions

- Definition. If $V_{12} > 0$, actions are strategic complements, if V_{12} actions are strategic substitutes.
- Definition. If $V_2 > 0$ actions have positive spillovers, if $V_2 < 0$ actions have negative spillovers.
- Definition. $\frac{\partial e_j(\bar{e};\theta)}{\partial\theta} = \frac{V_{1\theta}}{V_{11}}$ is the partial equilibrium response of e to θ, and $\frac{de}{d\theta} = \frac{V_{1\theta}}{V_{11}+V_{12}}$ is the general equilibrium response.
- Definition: Multiplier arises when general equilibrium response is in the same direction as partial equilibrium response, but greater.

Results 1

- Prop 1: if there are spillovers, then the SNE is inefficient
- Prop 2: If spillovers are positive, then the SNE levels of e are two low
- Prop 3: Strategic complementarites are necessary and sufficient for multipliers
- Note: When there are positive spillovers and strategic compl., may want to refer to such equilibrium outcome as exhibiting deficient demand

Results 2

- Prop 4: There can be multiple equilibrium. Strategic complementarities (and $\frac{-V_{12}}{V_{11}} > 1$) are necessary for multiple equilibrium.
- Prop 5: If positive spillovers and multiple equilibrium, then equilibrium are Pareto ranked with higher actions being better.
- Corr: If $e_j(\bar{e}) = \bar{e}$ over an interval, and there are positive spillovers, then there is a continuum of equilibrium increasing in e.

Relating to Walrasian Equilibrium

- A Walrasian equilibrium can be seen (setup) as a special case of the above game situation.
- In this case, we can see agents as choosing actions— expenditures or supply decisions – taking prices as given, but where prices depend on aggregate actions
- What differentiates the Walrasian equilibrium is at an equilibrium $V_2 = 0$, ie, there are no spillovers. This is not because agents are small, but because the effects of ones actions – at the margin– does not have a positive nor negative effect

- Consider a simple consumption labor supply problem, where the representative agent has preferences $U(c, e)$, and has the budget constraint $C = we + \frac{\Pi}{I}$, where $\frac{\Pi}{I}$ is a share of profits
- So we can see the agents problem as choosing e_j to solve $\max_I U(w(\bar{e}I)e_j + \frac{\Pi(\bar{e}I)}{I}, e_j)$.
- So want to think of $V(e, \bar{e}) = U(w(I \cdot \bar{e})e + \frac{\Pi(I \cdot \bar{e})}{I}, e)$
- The individual FOC is

$$wU_1 + U_2 = 0$$

Relating to Walrasian Equilibrium

- Since $V(e, \bar{e}) = U(w(I \cdot \bar{e})e + \frac{\Pi(I \cdot \bar{e})}{I}, e)$
- The social FOC (evaluate at the Walrasian equilibrium) is

$$wU_1 + U_2 + \sum_i U_1[\frac{\partial w}{\partial e}e + \frac{(F'(\bar{e}I) - w)\frac{\partial \bar{e}I}{\partial e}}{I} - \frac{\partial w}{\partial e}\frac{L}{I}] = 0$$

- which, because $F'(L) = w$, is equal to

$$wU_1 + U_2 + \sum_i U_1[\frac{\partial w}{\partial e}e + -\frac{\partial w}{\partial e}\frac{\bar{e}I}{I}] = 0$$

- re-written

$$wU_1 + U_2 + U_1[\frac{\partial w}{\partial e}\sum_i (e - \bar{e})] = 0$$

- which, by market clearing, equals to the individual level FOC.
- This is a case where actions have pecuniary effects, but these do not lead to inefficiencies

Relating to Walrasian Equilibrium
second example

- Endowment economy with two goods, endowment X_1 and X_1.
- Let e be purchase of good 1, and preference given by $U(e, \tilde{e})$, where $\tilde{e}$ is purchase of good 2. And let p be the relative price of good 1.
- $V(e, \bar{e}) = U(e, p(\bar{e}I)(X_1 - e) + X_2)$
- Then

$$\sum_i V_2 = \sum_i U_2(X_1 - e)\frac{\partial p}{\partial e} = 0$$

Relating to Global Games

- This simple interaction setup extends naturally to situation with imperfect information.
- For example, the θ in $V(e, \bar{e}; \theta)$ may be a latent variable, which is only observed with error by agents. This is the global games framework.
- Some of the insights from the global games setup
  1. Imperfect information may eliminate multiple equilibrium.
  2. Imperfect information affects the strength of complementarities (now reaction is a function of the expectation of the others actions, which depends on one's expectations of others expectation). This can cause expectation to play the role of a state variable
  3. This may allow for shocks to higher order beliefs (EX: in Angeletos and La'O)